ColoVision Analysis Report

Colorectal Cancer Segmentation Analysis

Report Date:	May 22, 2026 02:26 PM
Image File:	73.png
Analysis Type:	Binary Segmentation (ONNX Model)

Risk Assessment

Risk Level:	**High Risk**
Polyp Coverage:	5.07%
Model Confidence:	90.0%
Detected Pixels:	3,320
Total Pixels:	65,536

Clinical Recommendations

1. Immediate referral for colonoscopy with thorough polypectomy and histopathological assessment is recommended due to high-risk findings.

2. Schedule follow-up surveillance colonoscopy within 1 year to monitor for any new polyp formation or changes in existing lesions.

3. Consider genetic counseling and testing for hereditary colorectal cancer syndromes, given the high-risk classification.

4. Educate the patient on lifestyle modifications that may reduce colorectal cancer risk, including diet, exercise, and smoking cessation.

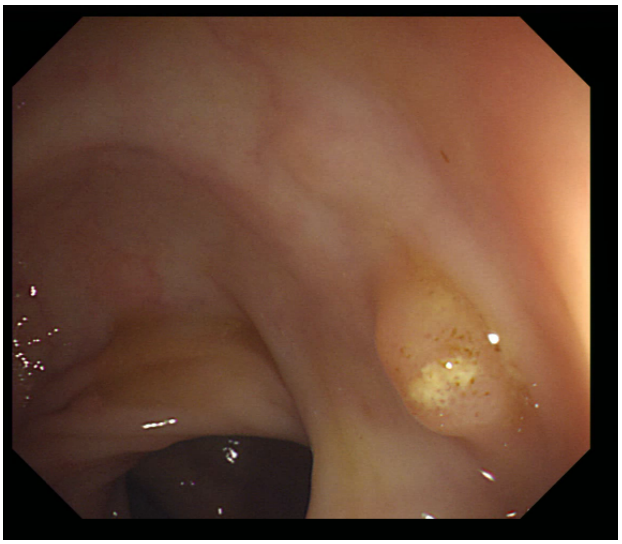
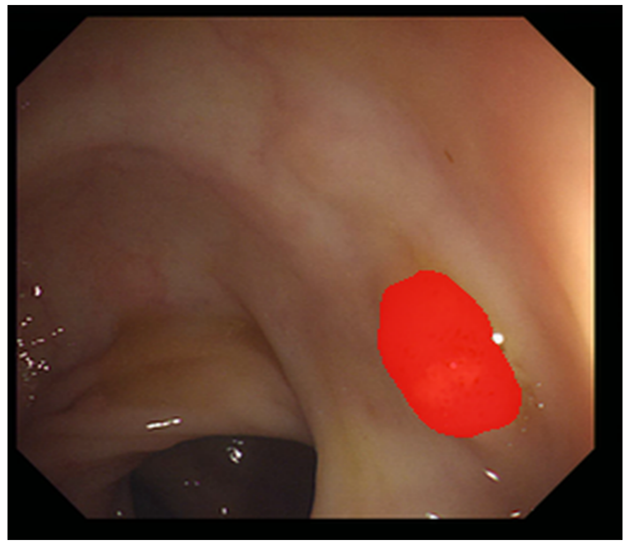
5. Document and discuss the importance of adherence to follow-up screening protocols with the patient to ensure timely surveillance.

Visual Analysis

Comparative analysis showing original colonoscopy image alongside AI segmentation results.

Original Image vs Segmentation Overlay

The left image shows the original colonoscopy view. The right image shows the AI-detected polyp regions highlighted in red, overlaid on the original image.

Original Colonoscopy Image	AI Segmentation Result
	
Unprocessed colonoscopy image	Red areas = Detected polyps

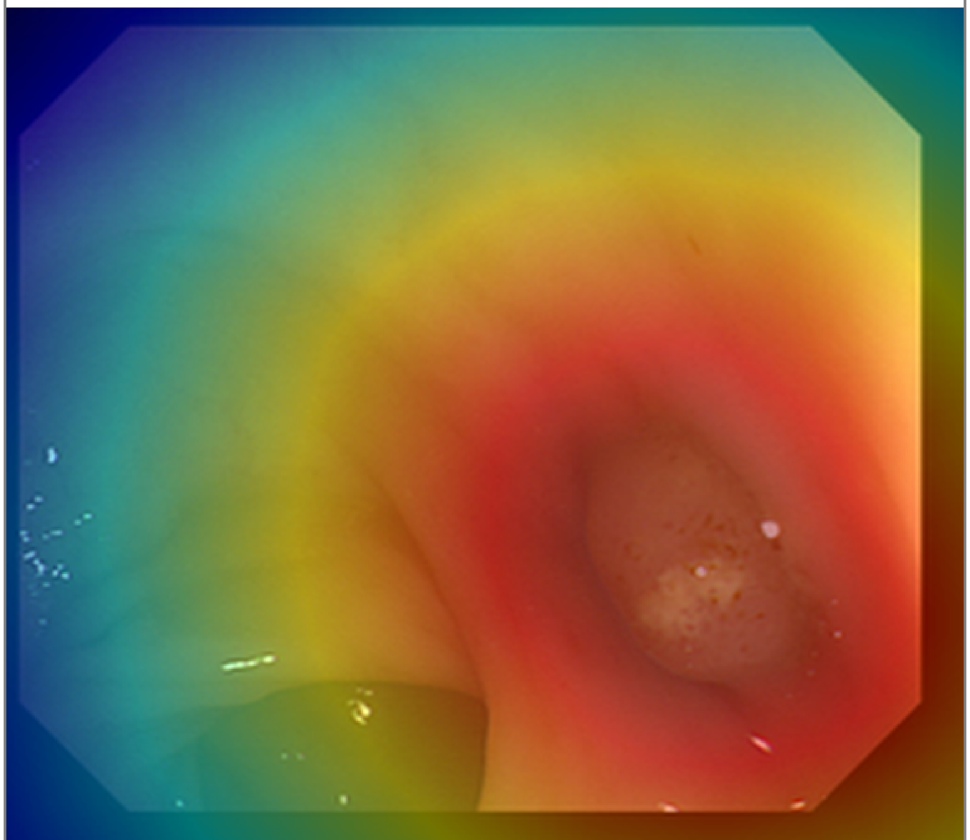
Segmentation Analysis Details

Aspect	Finding	Clinical Significance
Segmentation Mask	Binary detection of abnormal tissue regions	Identifies exact spatial location and extent of pathology
Coverage Area	3,320 pixels detected (5.07% of image)	Quantifies polyp size relative to field of view
Model Prediction	Confidence: 90.0% Risk Level: High Risk	Indicates reliability of automated detection

Grad-CAM Attention Heatmap

Gradient-weighted Class Activation Mapping (Grad-CAM) visualization showing where the neural network focused its attention. Red areas show high-attention polyp regions. Green areas indicate normal tissue where the model determined no pathology is present.

Model Attention Visualization



Red: Detected polyp regions (high confidence) | **Green:** Normal tissue (low risk)

Visualization Element	Interpretation	Medical Relevance
Red Overlay	Areas where the model detected polyp features	Primary regions requiring clinical examination
Green Overlay	Areas classified as normal healthy tissue	Regions with low pathology probability
Color Intensity	Indicates model confidence in its predictions	Stronger colors suggest higher certainty

Summary & Key Insights

Parameter	Value	Status
Polyp Coverage	5.07%	High
Total Pixels Analyzed	65,536	Complete
Abnormal Pixels	3,320	Detected
Model Confidence	90.0%	High
Risk Classification	High Risk	Critical

Model Information

Model Type	UNet with EfficientNet-B0 Backbone
Task	Binary Segmentation (Polyp Detection)
Format	ONNX Optimized
Input Size	256×256 pixels (RGB)
Output	Binary mask with probability scores

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